

Richard Qian Li

Creative technologist working on cities, simulation, modeling complex systems

I build software, websites, games and simulations on cities and systems. Before that, twenty years in architecture, ten of which as a director of design/product/operation at Vanke, one of China's largest property developers. Currently at NYU ITP, New York, at the intersection of art, architecture and technology.

ludwig.peking@gmail.com · GitHub · LinkedIn · YouTube · Portfolio · Sketches

Education

ITP, New York University 2025 – 2027 (exp.)
Master of Professional Studies, Interactive Telecommunications Program
Physical computing, computational media, 3D web and shader programming, live web, machine learning

University of Michigan, Ann Arbor 2007 – 2008
Master of Urban Design, with distinction

Tsinghua University, Beijing 1998 – 2003
Bachelor of Architecture

Self-directed study in computer science 2021 –
Coding, computational geometry, algorithms, machine learning

Creative technology & software

Independent developer 2022 – present
Simulation, games, generative art, installation, tools

- Built a body of procedural generation and simulation of the built environment, an update of the pattern language of architecture and the city with the layers of transportation, real estate market and surveillance. See Work.
- Shipped two professional tools for real estate: **ParcBalance**, a parking-lot pricing simulator that models the urban context, customer behaviour and spatial distribution of lots; and **Decisive**, a SketchUp plug-in that couples design/product schemes to real-time cash-flow calculation (Ruby API).
- Games, installations and spatial works at ITP.

Exhibitions

- ITP/IMA Winter Show 2025, New York University: Hexagonal World, a projected sand-table city simulator.
- ITP Spring Show 2026, New York University: the Algorithms of the City.

Career

Vanke Co., Beijing / Shenyang / Taiyuan 2011 – 2025
Director of Design, Chief Architect, Senior Partner, Branch Director
Led the design of large mixed-use, residential and industrial-park developments, and ran the regional design team. Selected projects are in the architecture archive.

EE&K Architects, Washington, DC 2008 – 2011
Architect, Project Architect. Urban design for Qingdao Waterfront, Mount Vernon Square and North Capitol Street.

Obermeyer Planen + Beraten GmbH, Beijing / Munich 2003 – 2006
Architect, Project Architect

Teaching, lectures & writing

Jury member and design critic 2019 – 2021
Tsinghua University; Shenyang Jianzhu University

Guest lectures 2022
Beijing Jiaotong University

Lecturer and trainer 2011 – 2025
Vanke Co. in-house design training, Beijing and Shenyang

An Evolutionary History of the City 2018 – present
Book in progress, 38 chapters published online
Read online

Banal Smasher (陳腐粉碎機), podcast on urban history 2016 – present
120 episodes in Mandarin, about one million accumulated listens
Listen on Ximalaya

Algorithms of the City 2026
Video essay on procedural models of urban growth

100 Mysteries of Science Explained 2017
New Star Press, ISBN 9787513313063. Translation co-editor.

Awards

Adobe x NYU Competition, Best Branding 2026
Altrovia: identity and site design for a clinical-trial design company, with Minglu Zhong, Tiffany Ziyi Yuan and Kurt Mu Qian

Parsons New School Creative Center Award 2026
WebXR Playful NYC: a multiplayer WebXR scene of a 50-square-foot New York room shared by two roommates, with Elizabeth Kezia Widjaja, Aditya De, Eros Carrasco

Melbourne Design Awards, Gold 2022
Shenyang Sino-Japanese Industrial Park, as director of the design team (Vanke with Topscape)

AIA New York Chapter, Urban Design Honor Award 2013
Qingdao Waterfront, team member at EE&K

Youth STEM Elite of China, nomination 1997
Fifteen nominees per province

Skills

Languages: JavaScript, Python, GLSL / HLSL, C#, Ruby (SketchUp API)

Creative coding & graphics: p5.js, three.js, React Three Fiber, WebGL shaders, Unity, Blender and bpy, projection mapping, Gaussian splatting

Simulation & algorithms: Agent-based simulation, computational geometry, genetic algorithms

Machine learning & vision: scikit-learn, ml5, MediaPipe, OpenCV, Azure Kinect, LLM tooling

Physical & spatial: microcontroller and sensors, projection mapping and calibration, digital fabrication

Web: Node, Express, WebSockets, D3, static sites

Design: AutoCAD, Revit, SketchUp, 3ds Max, Adobe Creative Suite, hand drawing

Spoken languages: Mandarin (native), English (fluent), Japanese (basic)